Chih-Wei Joshua Liu

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EDUCATION

Stanford University — Stanford, CA, USA
Biophysics PhD Program · **Thesis advisor:** Grant M. Rotskoff · **GPA:** 4.0/4.0 — 09/2024 – present
Classes: PHYSICS212 Statistical Mechanics, MATH220A Partial Differential Equations, CS229 Machine Learning

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
Bachelor of Science · **Major:** Physics · **GPA:** 4.9/5.0 · **MCAT:** 528/528 — 05/2022
Classes: 18.354 Nonlinear Dynamics, 5.353 Spectroscopy · **Honors:** Joel Matthew Orloff Award, MIT News profile

National Experimental High School — Hsinchu, Taiwan, ROC
USABO: Semifinalist (2015 – 2018); High Honorist (2017) · **ACT:** 36/36 · **SAT:** 1600/1600 — 05/2018

EXPERIENCE

MIT Physics of Living and Non-Equilibrium Systems Group — 06/2022 – 08/2024
Research Assistant — Cambridge, MA, USA
Implemented soft-matter experiments and developed polyspectral, ensemble-learning, and neural-network analyses.

MIT Department of Physics — 02/2024 – 05/2024
Teaching Assistant, 8.241 Introduction to Biological Physics — Cambridge, MA, USA
Led recitations on diffusions, FD theorems, and nonequilibrium phase transitions. Assisted optical-trap experiments.

Massachusetts General Hospital — 06/2021 – 08/2021
Data Science Intern — Boston, MA, USA
Implemented GLS models of patient data and authored two clinical publications using `R`, `SPSS`, and `statsmodels`.

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, **Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data**, *Phys. Rev. X*, provisionally accepted (2026). [code]

- Derived the first maximum-likelihood parameterization of non-Markovian master equations satisfying all physical constraints in general classical-stochastic processes. **Skills:** convex optimization, `lightning`
- Validated likelihood-maximization algorithms on protein experiments. Set SOTA in benchmark extraction of HP35 protein-folding kinetics from atomistic molecular-dynamics simulations. **Skills:** autoregressive models, `mdtraj`

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [code]

- Developed variational-autoencoder representations of complex-Ginzburg-Landau dynamics to measure entropy production by active-gel excitations in simulated and experimental cell cortices. **Skills:** `pillow`, `torch`, `torchvision`

A. Yona, **C.-W.J. Liu**, et al., and J. Gore, **Adaptive evolution identifies candidate genetic routes associated with oxalate utilization in probiotic and gut-derived *E. coli***, *Cell Rep. Methods* **7**, 101644 (2026).

- Implemented parallelized simulations of high-dimensional generalized-Lotka-Volterra equations to delineate attractor basins in gut-microbe community assembly. **Skills:** `pymol`, `DynamicalSystems.jl`, `SciML.jl`

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution. Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

- Trained classifier to distinguish particle classes with nonreciprocal interactions, enabling analyses of previously experimentally unobserved nonequilibrium states. **Skills:** `scikit-learn`, `scipy`, `numpy`, `pandas`, `opencv`

GRANTS AND LEADERSHIP

National Science Foundation Graduate Research Fellowship (GRFP, 2024, Physics)
National Defense Science and Engineering Graduate Fellowship (2024, Physics, declined for GRFP)
MIT Society of Physics Students and Sigma Pi Sigma: honor-society inductee (2022); President (2022)
Phi Kappa Theta – Massachusetts Eta: Pledgemaster (2021); sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage as Service Chair (2019–2020).

SKILLS AND INTERESTS

Laboratory: NMR spectroscopy, fluorescence microscopy · **Computing:** `jax`, `pydantic`, `uv`, `ty`, `ruff`, `slurm`
Natural language: Standard Chinese (native) · **Personal interests:** phonology, archaeology, ichthyology, Set
Research interests: nonequilibrium statistical physics, optimal transport, time series, LLM expressivity, SAT solvers